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NPS/002/012 Technical Specification for Low Voltage Underground Link Boxes

1 Purpose

The purpose of this document is to detail the requirements of CE Electric UK in relation to underground disconnecting link boxes, which will be utilised on the company's low voltage distribution network.

This document supersedes the following documents, all copies of which should be destroyed.

Ref.	Version	Title
NPS/002/012	1 (June 06)	Technical Specification for Low Voltage Underground Link Boxes

2 Scope

This document details the technical requirements of CE-Electric UK in relation to 2 and 4-way resin filled underground link disconnecting boxes. The proposed link boxes shall include pit frame; concrete filled cover and links and shall match the current rating of the largest cable size.

Link boxes will generally comply with testing requirement of ENATS C81- 4, BS EN 50393 and BS 7888.

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3 Requirements

3.1 Link Box Shell

The split joint shell shall be suitable for use with an appropriate filling medium, either polyurethane or acrylic based and shall be a minimum 2mm thick. The lower section shall be flat bottomed to provide a positive location onto a flat slab installed in the proposed joint bay by the installer. The upper half of the shell shall be clear; this is to provide verification of cable core clearances etc. The shell shall be capable of accepting 3 or 4 core cables in the range 95mm² - 300mm² to BS7870, Part 3, Section 3.40.

3.2 Turret Assembly

A factory-assembled turret which positively fits into the lower half of the joint shell shall be capable of accommodating either solid links or fuse carrier, operated by wedge tightening thumb screws which will incorporate a J type fuse link with 83mm centres as per BS88-5. Links shall be supplied for each cable way.

Cable terminations/contacts shall be manufactured from tinned hard drawn copper and capable of carrying the rated current of the largest cable to be terminated.

Adequate phase to phase and phase to neutral/earth segregation shall be provided with suitable insulating barriers as appropriate. This barrier will also be sufficient to stop electrical tracking between links caused by condensation.

Provision of suitable neutral and earth connection facilities for testing and fault location shall be provided within the turret assembly.

Range taking phase and neutral connectors to meet the requirements of Engineering Recommendation C79-1 and BS EN 61238: Pt 1, capable of accepting stranded copper or solid aluminium conductors in the range 185mm² to 300mm², shall be factory fitted and utilise shear head technology.

Plastic removable shrouds shall be provided for covering unused contacts.

3.3 Bell Cover

The 2 and 4-way link box shall be supplied with an insulated bell cover, and shall be so designed to prevent water entering the link chamber when submerged to a minimum depth of 1metre head of water. This shall incorporate a notice indicating "Danger Electric" or equivalent Danger of Death Sign to BS5499:5.

3.4 Pit Assembly

The 2 & 4-way link boxes shall be supplied with either a GRP or HDPE housing. This will sit over the link box and locate on it so that the bell cover is central. It will prevent the ingress of any backfill material to the link box.

The assembly shall be horizontally sectionalised to enable sections to be added or removed, depending on the depth of the link box to finished ground level. The pit frame shall meet the test load requirements of BS EN 124 class B125 when installed.

3.5 Pavement Access Cover & Frame

The frame for the pavement access cover, which shall fit onto the top of the pit assembly shall be fabricated from steel and hot dipped galvanised. The frame shall be complete with integral "Skirt" to prohibit the ingress of backfill material into the pavement chamber. The frame shall incorporate an

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adjustment system that allows up to approximately 100mm vertical movement of the cover. The adjustment shall only be achieved from inside the pit with the pavement access cover removed.

The pavement cover for the 2-way link box assembly shall be a single galvanised steel tray, concrete filled with 2 x "keyhole" type slots to aid removal from the frame. This shall incorporate a notice indicating "Danger Electric" or equivalent Danger of Death Sign to BS5499:5.

The steel frame for the 4-way link box shall incorporate removable galvanised steel cross-member to allow 2 x single pavement covers (as detailed above) to be installed.

The access cover and frame shall meet the test load requirements of BS EN124 class B125 when installed.

There is also a requirement for an un-filled pavement cover. This cover will be utilised when a link box is to be installed in a block paved area etc.

3.6 Circuit Identification

Circuit identification label(s) of an appropriate size, manufactured from insulating material and suitable for engraving shall be incorporated into the link box, below the bell cover for each cable way. Labels shall be positively located and easy to remove.

3.7 Tool Box/Container

The link box shall include an insulated waterproof container for the safe keeping of removable links and plastic shrouds. The container shall be either located on top of or under the bell housing.

3.8 Extras

The tool box shall also include a suitable tube of contact grease (or similar) for lubricating the jaws of the links to aid installation.

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4 References

4.1 External Documentation

All equipment covered by this specification shall be in accordance with the latest versions of ENATS, BSEN, IEC and other relevant standards except where varied by this specification.

In this respect the following documents are particularly relevant:-

Reference	Title
C81- 4	Type tests for joints for 600/1000 volt CNE cable systems.
BS EN 61238	Compression and Mechanical for Power Cables up to 36kV.
BS EN 50393	Test Methods and Requirements for Accessories for use on Distribution Cables of rated voltage 0.6/1.0 (1.2) kV.
BS 7888	LV and MV Accessories for Power Cables.
BS EN 60497	Low Voltage Switchgear and Control Gear.
BS 88 – 5	Cartridge fuses for voltages up to and including 1000 V ac. and 1500 V dc. Specification of supplementary requirements for fuse-links for use in ac. electricity supply networks.
C79	Type approval tests for connectors and terminations for Aluminium conductors of insulated power cables.
BS -EN 124 - 1994 (Class B125)	Gully tops and manhole tops for vehicular and pedestrian areas. Design requirements, type testing, marking and quality control.
BS 7870 Part 3, Section 3.40	LV and MV Polymeric Insulated Cables for use by Distribution and Generating Utilities -
BS 5499-5	Graphical symbols and signs. Safety signs, including fire safety signs. Signs with specific safety meanings

4.2 Internal documentation

Reference	Title

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4.3 Amendments from Previous Version

Clause	Subject	Amendments
1	Purpose	Previous versions table amended.
2	Scope	Reference to external documents updated.
2.1	Contents	Table of contents added.
3.1	Joint Shell	Reference made to acrylic based filling medium and 4c cables.
3.3 and 3.5	Bell Cover, Pavement Access Cover & Frame	Amended to included requirement to incorporate a notice indicating "Danger Electric" or equivalent Danger of Death Sign.
3.6	Circuit Identification	Minor amendments to wording. Requirement for a tool to be used to take out labels removed.
4.1	External Documents	Opening paragraph amended to detail requirement to comply with latest versions of ENATS, IEC, BS and other relevant standards. References updated to include BS EN 61238, BS EN 50393, BS EN 60497, C81-4, BS7870 Pt 3, Section 3.40 and BS5499 Pt 5.
6	Authority for Issue	Amended to reflect new business structure.
Appx 4	Self Certification Conformance Declaration	Tables added.

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5 Definitions

Term	Definition
GRP	Glass Reinforced Plastic
HDPE	High Density Polyethylene

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6 Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
V Gorny	Standards Engineer	V Gorny	17/06/2010

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
C Holdsworth	Standards and Assessment Manager	C Holdsworth	22/06/2010

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
R McMahon	Standards Engineer	R McMahon	21/06/2010

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S Johnson	DBD Administrator	S Johnson	27/07/2010

6.5 Approval

Approval is given for the content of this document

		Sign	Date
C Holdsworth	Standards and Assessment Manager	C Holdsworth	22/06/2010

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
M Nicholson	Head of System Strategy	M Nicholson	13/07/2010

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Appendix 1 - Requirements

Description	Catalogue Number
2-Way 3c Underground Link Box	056077
4-Way 3c Underground Link Box	056168
2-Way concrete Replacement Pavement Cover	056291
Un-filled 2-Way Pavement Cover	TBA

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Appendix 2 - Addendum to Supplier Requirement

Supporting evidence of compliance with type tests shall be submitted with the completed tender document.

Manufacturers may provide alternative tenders for items not complying with the above specification. This shall be clearly stated together with detailed descriptions of any variation from the specification, together with drawings and test results.

The supplier shall provide with the tender full technical details of the equipment offered and shall indicate any divergence from these standards or specifications.

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Appendix 3 - Logistical requirements

To enable CE Electric UK to store the product(s) in accordance with the manufacturer's recommendations the Tenderer should provide details of the recommended storage environment with respect to each tendered product.

Details should be provided where relevant in respect to the minimum and maximum exposure levels, frequency of exposure and duration of exposure of the packaged item with respect to;

- | | | | |
|---|---------------------|---|-----------------------|
| * | Ambient temperature | * | Atmospheric corrosion |
| * | Humidity | * | Impact |
| * | Water | * | Vibration |
| * | Dust | * | Solar radiation |

The Tenderer shall ensure that each item is suitably packaged and protection to maintain the product and packaging as "fit for service" prior to installation taking account of the potential for an outdoor storage environment. All packaging shall be sufficiently durable giving regard to the function, reasonable use and contents of the packaging. Where product packages tendered are made up of sub packages all the sub packages shall unless varied by this specification, be supplied securely packaged together. Where items are provided in bagged/boxed form the material from which the bags are manufactured shall be capable of sustaining the package weight and resisting puncture by the materials within. The Tenderer shall submit at the time of tendering the details of the proposed packaging (i.e. materials composition and structure) to be used for each product. Where the Tenderer is unable to provide packaging suitable for outdoor storage then this should be stated at the time of tender.

Palletised goods shall be supplied on standard 1200mm x 1000mm pallets.

Clearly legible, easily identifiable, durable and unambiguous labelling shall be applied to each individual and where relevant multiple package of like products. Where products packages tendered are made up of sub packages each sub packages shall be marked. As a minimum requirement the following shall be included;

- * Manufacturer's trademark or name
- * Supplier's trademark or name
- * Description of item
- * Date of packaging and/or batch number
- * CE Electric UK product code
- * Weight

The Tenderer shall submit at the time of tendering a sample of the proposed labelling for each product package type.

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Appendix 4 - SELF CERTIFICATION CONFORMANCE DECLARATION

The Technical Specification for Low Voltage Underground Link Boxes shall comply with the latest issues of the relevant national and international standards, including ENATS C81/4, BS EN 61238, BS 7888 and BS EN 60497.

Additionally this technical specification is intended to amplify and/or clarify requirements relating to these Standards.

This self declaration sheet identifies the clauses of the aforementioned standards relevant to The Technical Specification for Replacement Outdoor Meter Cupboard Frame and Doors for use on the CE Electric UK distribution network. The manufacturer shall declare conformance or otherwise, clause by clause, using the following levels of conformance declaration codes.

Conformance declaration codes

N/A = Clause is not applicable/ appropriate to the product

Cs1 = The product conforms fully with the requirements of this clause

Cs2 = The product conforms partially with the requirements of this clause

Cs3 = The product does not conform to the requirements of this clause

Cs4 = The product does not currently conform to the requirements of this clause, but the manufacturer proposes to modify and test the product in order to conform.

Instructions for completion

- When Cs1 code is entered no remark is necessary
- When any other code is entered the reason for non- conformance shall be entered
- Prefix each remark with the relevant 'BS EN' 'IEC' or 'ENATS' as appropriate.

Manufacturer:

Product Reference:

Name:

Signature:

Date:

NOTE: One sheet shall be completed for each item or variant submitted.

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Appendix 4 Cont'**Technical Specification for Low Voltage Underground Link Boxes**

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	Requirement	Conformance Code	Remarks
General	Available in 2 way or 4 way options.		
3.1 Link Box Shell	Suitable for cable sizes 95mm ² – 300mm ² .		
	Suitable for either polyurethane or acrylic based resins.		
	Flat bottomed, horizontally split shell. Min thickness 2mm, upper part clear to enable clearances to be checked.		
3.2 Turret Assembly	Positively located with appropriate mechanical and heat resisting properties.		
	Hard draw tinned copper contacts, suitable for links or J type fuse link with 83mm centres to BS88-5. Current rated for largest cable size.		
	Integral connectors suitable for stranded copper or solid aluminium conductors in the range 185mm ² to 300mm ² , tested to Engineering Recommendation C79-1 / BS EN 61238.		
	Links provided (for all cable ways).		
	Inter- Phase Barriers.		
	Neutral / Earth Connection Access (testing / fault location / generator connection).		
	Shrouds for unused cable ways.		

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Appendix 4 Cont'

• Resin Encapsulated Link Box to NPS/002/012			
	Requirement	Conformance Code	Remarks
3.3 Bell Cover	Insulated, suitable to prevent water ingress.		
	Incorporates a notice indicating "Danger Electric" or equivalent Danger of Death Sign to BS5499:5.		
3.4 Pit Assembly	GRP or HDPE housing, to provided access to and prevent backfill material covering link box assembly.		
	Horizontally stackable sections to enable adjustment for depth of link box assembly.		
	Suitable for access frame and cover 3.5		
3.5 Pavement Access Frame and Cover	Frame manufactured from hot dipped galvanised steel, with adjustable "skirt" to allow 100mm of vertical movement and prevent ingress of backfill material link box assembly. .		
	Covers shall comprise a galvanised steel tray, concrete filled with 2 x "keyhole" type slots to aid removal from the frame:- 1 x cover / 2 way link box. 2 x covers / 2 way link box.		
	Load Test Requirements Pit assembly, access frame and covers shall meet the load requirements of BS EN124 class B125.		
	Incorporates a notice indicating "Danger Electric" or equivalent Danger of Death Sign to BS5499:5.		
3.6 Circuit Labels	Non conductive, appropriately sized, suitable for engraving, positively located and easy to remove without tools.		
3.7 Tool Box	Waterproof container suitable for storing unused links, contact grease and shrouds.		
3.8 Extra's	Contact Grease		

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Appendix 4 Cont'

• Resin Encapsulated Link Box to NPS/002/012 (Tests to requirements of C81/4, BS EN 50393, unless specified otherwise)			
	Requirement	Conformance Code	Remarks
Connectors	Range taking for cable sizes 95mm ² – 300mm ² , Tested to C79 / BS EN 61238: Pt 1		
Live Make / Break Tests	BS EN 60947-1: 2004 sections 7.2.5 & 8.3.3.5		
Temperature Rise	BS EN 60947-1: 2004 sections 7.2.2 & 8.3.3.3 :- Connectors: Max Temp 65°C Shell and Resin: Max Temp 25°C Bell Cover: Max Temp 15°C		
Impact Test	BS EN50393:2006 section 8.5 (2m height)		
4kV AC Voltage Withstand Test	BS EN 50393:2006 section 8.3		
1kV DC Insulation Resistance Test	50MΩ or greater		
4kV AC Voltage Withstand Test	BS EN 50393:2006 section 8.3 (Inclined 10° under water; min 3hrs / 2m head)		
1kV DC Insulation Resistance Test	50MΩ or greater (Inclined 10° under water; min 3hrs / 2m head)		
Sheath damage and Electrical Heat Cycling	2m height. BS EN50393:2006 section 8.6 (Inclined 10° under water; min 3hrs / 2m head)		
4kV AC Voltage Withstand Test	BS EN50393:2006 section 8.3 (under water, 2m head)		
1kV DC Insulation Resistance Test	50MΩ or greater (under water, 2m head)		
Bell Cover DC Withstand	BS EN 50393 section 8.4 or BS 923-1 1990 clause 14.1		
Visual Inspection	Examination for water ingress / corrosion BS EN 50393:2006 section 8.8		